

③ 2) defining $\Gamma = 2.354\sigma$

Starting w/definition of Γ

at two points where the curve is half its maximum
 2 points half-max.

$$P_G\left(M \pm \frac{\Gamma}{2}, M, \sigma\right) = \frac{1}{2} P_G(M, M, \sigma)$$

knowing that the probability density function of a Gaussian is given by

$$P_G(x; M, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-M)^2}{2\sigma^2}\right)$$

at the Gaussian's peak $x=M$, so $P_G(M; M, \sigma)$ is given by

$$P_G(x=M, M, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(M-M)^2}{2\sigma^2}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp(0) = \frac{1}{\sigma\sqrt{2\pi}} = P_G(M, M, \sigma)$$

Substituting back into the definition of Γ , we now know that

$$P_G\left(M \pm \frac{\Gamma}{2}, M, \sigma\right) = \frac{1}{2} P_G(M, M, \sigma) = \frac{1}{2\sigma\sqrt{2\pi}}$$

Now to resolve the left side and finding the probability density function for a randomly observed value, when that observed value is the mean plus or minus half of the half-width, $x = M \pm \frac{\Gamma}{2}$

$$P_G\left(x = M \pm \frac{\Gamma}{2}, M, \sigma\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\left(M \pm \frac{\Gamma}{2}\right) - M\right)^2}{2\sigma^2}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\frac{\Gamma}{2}\right)^2}{2\sigma^2}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\Gamma^2}{8\sigma^2}\right)$$

Substituting back into the definition of Γ

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\Gamma^2}{8\sigma^2}\right) = P_G\left(M \pm \frac{\Gamma}{2}, M, \sigma\right) = \frac{1}{2} P_G(M, M, \sigma) = \frac{1}{2\sigma\sqrt{2\pi}}$$

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\Gamma^2}{8\sigma^2}\right) = \frac{1}{2\sigma\sqrt{2\pi}}$$

$$\exp\left(-\frac{\Gamma^2}{8\sigma^2}\right) = \frac{1}{2} \rightarrow -\frac{\Gamma^2}{8\sigma^2} = \ln\left(\frac{1}{2}\right)$$

$$* \ln\left(\frac{1}{2}\right) = \ln(2^{-1}) = -\ln(2)$$

$$\rightarrow -\frac{\Gamma^2}{8\sigma^2} = -\ln(2) \rightarrow \frac{\Gamma^2}{8\sigma^2} = \ln(2) \rightarrow \Gamma = \sqrt{8\sigma^2 \ln(2)} = \sigma \sqrt{8 \ln(2)} = 2.354\sigma$$

$$\boxed{\Gamma = 2.354\sigma}$$